

(PROJECT PROPOSAL)



FARMING DATA COLLECTOR BY SENSOR TECHNOLOGY

MANINTORN NANTAWANOTAYAN

WATTHANACHAI PHAKDEE

SANDUSIT BUNLERT

SIRICHAIR PHUKRONGHIN

LI LI

BACHELOR OF ENGINEERING

IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

2021

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

FARMING DATA COLLECTOR BY SENSOR TECHNOLOGY

MANINTORN NANTAWANOTAYAN

WATTHANACHAI PHAKDEE

SANDUSIT BUNLERT

SIRICHAIR PHUKRONGHIN

LI LI

A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

2021

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

FARMING DATA COLLECTOR BY SENSOR TECHNOLOGY

MANINTORN NANTAWANOTAYAN

WATTHANACHAI PHAKDEE

SANDUSIT BUNLERT

SIRICHAIR PHUKRONGHIN

LI LI

THIS COMPUTER ENGINEERING PROJECT HAS BEEN APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING 2021

EXAMINING COMMITTEE

..... ADVISOR

(Asst. Prof. Gp. Capt. Dr. Thongchai Yooyatiwong)

..... COMMITTEE

(Aj. Dr. Chayapol Kamyod)

..... COMMITTEE

(Aj. Paweena Suebsombut)

ACKNOWLEDGEMENT

This project will not be successful without the kind support of the advisor, committees and all supportive persons. We would like to thank everyone for their advice and help. We are also grateful to our parents and friends for their continuous encouragement throughout this project. This accomplishment would not have been possible without them. Thank you.

MANINTORN NANTAWANOTAYAN

WATTHANACHAI PHAKDEE

SANDUSIT BUNLERT

SIRICHAIR PHUKRONGHIN

LI LI

Title Farming Data Collector By Sensor Technology

Author Mr. Manintorn Nantawanotayan

Mr. Watthanachai Phakdee

Mr. Sandusit Bunlert

Mr. Sirichai Phukronghin

Miss. Li Li

Degree Bachelor of Engineering

(Computer Engineering)

Supervisory Committee	Asst. Prof. Gp. Capt. Dr. Thongchai Yooyatiwong	Advisor
	Aj. Dr. Chayapol Kamyod	Committee
	Aj. Paweena Suebsombut	Committee

Abstract

The Project's purpose is for studying and developing a data collector system from sensors that are used in Smart Farm. ESP 32 gets command from Arduino IDE, sensors will collect environmental data, camera will collect pictures of crops, then upload data and pictures to MySQL cloud database. The administrators can check if there is corrupted data and do data cleaning. The data will show into the dashboard on a web application. The data will store and wait to be used in further benefits for example, use in data analysis or training AI.

Keyword: Agriculture, Database, Data Collector, IoT Sensor, Smart Farm

CONTENTS

ACKNOWLEDGEMENT	iii
Abstract	iv
LIST OF TABLES	vi
LIST OF FIGURES	vii
CHAPTER 1 INTRODUCTION	1
1.1 Background and Rationale	
1.2 Objective	
1.3 Scope	
1.4 Methodology	
1.5 Plan	
1.6 Expected Result	
1.7 Place and Equipment	
CHAPTER 2 LITERATURE REVIEW	
2.1 Related Theory	
2.2 Related Work	
2.3 Related Technology	
CHAPTER 3 ANALYSIS AND DESIGN	
3.1 Existing System	
3.2 New System	
3.3 Use case diagram	
3.4 Activity diagram	
CHAPTER 4 CONCLUSION	
4.1 Conclusion	
4.2 Discussion	
REFERENCES	

LIST OF TABLES

Table	Page
Table 1.1 Working plan	2
Table 2.1 Comparing MySQL vs Firebase vs MongoDB	
Table 2.2 Comparing MySQL vs MongoDB	
Table 2.3 Comparing MySQL vs Firebase	

LIST OF FIGURES

Figure	Page
Figure 1.1 Conceptual Diagram	
Figure 2.1 Cycle of Smart Farming	
Figure 2.2 Auto Weather Station	
Figure 2.3 Arduino Uno Interface	
Figure 2.4 Go or Golang, Programming language	
Figure 2.5 Example of Grafana Dashboard	
Figure 2.6 DHT11 Picture	
Figure 2.7 ESP32-CAM and OV2640 Picture	
Figure 2.8 OV2640 Picture	
Figure 2.9 Capacitive Soil Moisture Sensor Picture	
Figure 2.10 NPK and pH Soil Sensor	
Figure 3.1 Design Draft	
Figure 3.2	
Figure 3.3	
Figure 3.4	
Figure 3.5	

Chapter 1

INTRODUCTION

1.1 Background and Rationale

Nowadays, Technology information and data collection has been used increasingly in daily life. It is widely used in all occupations, especially agriculture, which has been important with our country for a long time. Agriculture farmers still face many problems with effective cultivation and effective products due to lack of agricultural knowledge and in-depth information for deciding what environment is good for the plants. The example of IoT in agriculture is “Smart Farm”, which applies technology to farms and helps farmers to be more convenient and save their time. Smart Farm has sensors to collect data such as temperature, humidity, and more, making farmers can view data from sensors in their farm from far distances.

In this project, we want to collect environmental data and pictures of crops to be used in further benefits such as, data analysis for those who are interested in finding in-depth analysis of the relationship between environmental data in agriculture area and growth rate of crops, use for machine learning, training AI, and more.

1.2 Objective

- 1.2.1 Improve system to collect data from sensors in agriculture areas.
- 1.2.2 Improve the data collector system, store and share the data for further use.
- 1.2.3 Improve the convenience to read the data from the sensor on the dashboard.

1.3 Scope

Studying and developing a system for collecting environmental data and crop data in agriculture areas. The data will be stored in a cloud database for further use, and farmers can view the dashboard on a web application. The crop that we choose in this project is Raspberry.

- 1.3.1 System will collect the data in agriculture area, which are
 - 1.3.1.1 Soil Humidity, Soil pH, and Soil NPK
 - 1.3.1.2 Air Humidity and Air Temperature
 - 1.3.1.3 Pictures of raspberry from camera
- 1.3.2 Data and pictures will be sent to a MySQL cloud database.
- 1.3.3 After data is sent to the cloud database, the administrator will do data cleaning if there is any corrupted data.
- 1.3.4 Farmers can view the data on the web application, Grafana.

1.3.5 System Design Picture

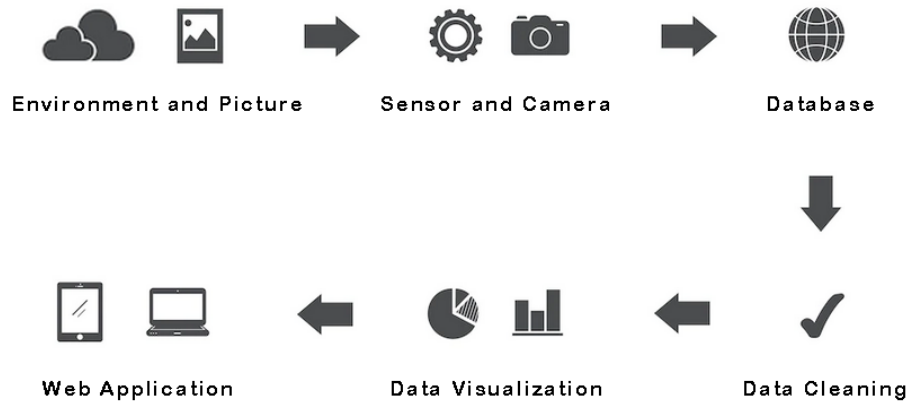


Figure 1.1 Conceptual Diagram

1.4 Methodology

- 1.4.1 Proposal preparation and defense
- 1.4.2 Literature review and Requirement gathering
- 1.4.3 System Analysis
- 1.4.4 System Design
- 1.4.5 Pre-Project document preparation
- 1.4.6 Pre-project Presentation
- 1.4.7 System Development
- 1.4.8 System Testing
- 1.4.9 Project Presentation

1.5 Plan

Table 1.1 Working plan

	Pre-project					Project				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
Learning System Analysis and System Design										
Pre-Project presentation										
System development										
System testing										
Project presentation										

Table 1.1 Table Plan

1.6 Expect Result

1.6.1 The System can collect data efficiently and accurately.

1.6.2 The System can store data on the database and can share for further use.

1.6.3 Users can view information on the web application conveniently.

1.7 Place and Equipment

1.7.1 Place: Raspberry Farm

1.7.2 Hardware :

1) ESP32-CAM and OV2640	220.-
2) BH1750FVI Light Intensity	120.-
3) Capacitive Soil Moisture Sensor	45.-
4) Soil NPK Sensor	450.-
6) DHT11	45.-
7) Battery	
TOTAL	835.-

Chapter 2

LITERATURE REVIEW

2.1 Related Theory

2.1.1 Big data in Agriculture

Big data in Agriculture is used with other technologies such as sensors in IoT, data analysis and more. It can help many things such as, to count and observe demand of food, collect environment data, observe if climate changes and more. To get big data, start with IoT devices for data collection. Sensors collect real-time data from the climate in agriculture areas, soil, air, light, and plants. Then do the data analysis on these large amounts of data collected from sensors, or combined with other data from other sources, upload them to a cloud database. Lastly, find patterns or relationships to pinpoint the problems, possible to see if there are anomalies or observe to predict problems before they occur.

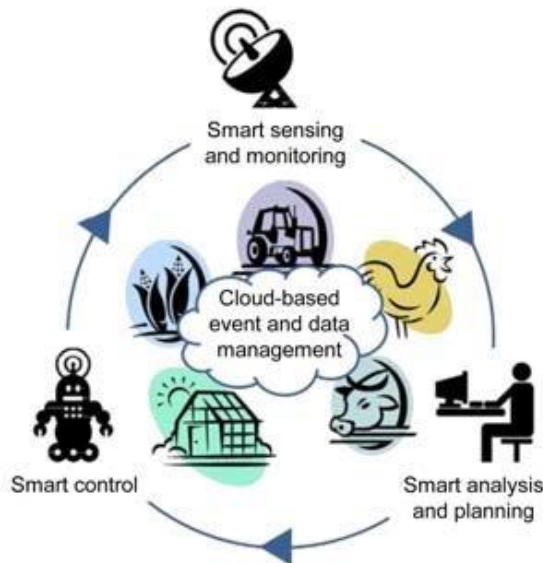


Figure 2.1 Cycle in Smart Farming with cloud-based event and data management [1]

2.2 Related Work

2.2.1 Arduino Greenhouse [2]

Arduino Greenhouse, or related work, is using sensors to detect soil humidity, and air temperature and control devices in many situations to make greenhouse climates good for plants. If humidity is too low, the system will control the hydroponic system to spray out water for plants. If the temperature is high, open the fan to make it cooler, If the temperature is low, open the infrared bulb to make it warmer.

2.2.2 Auto Weather Station for Agriculture [3]

Automatic Weather Station use in agriculture for weather researching. It has many high quality sensors, can measure many climates and environments such as, wind speed, wind direction, air temperature, humidity, rainfall, barometric pressure, soil temperature, soil moisture, and other parameters. It has high-reliability sensors for data collection, has software filters to do data cleaning, and removes noise and errors. The data can be sent both on wired or wireless. It has batteries which are charged from solar cell panels and batteries can store energy for around 20 days at maximum charging if the setting is to send data to the cloud database at every hour.

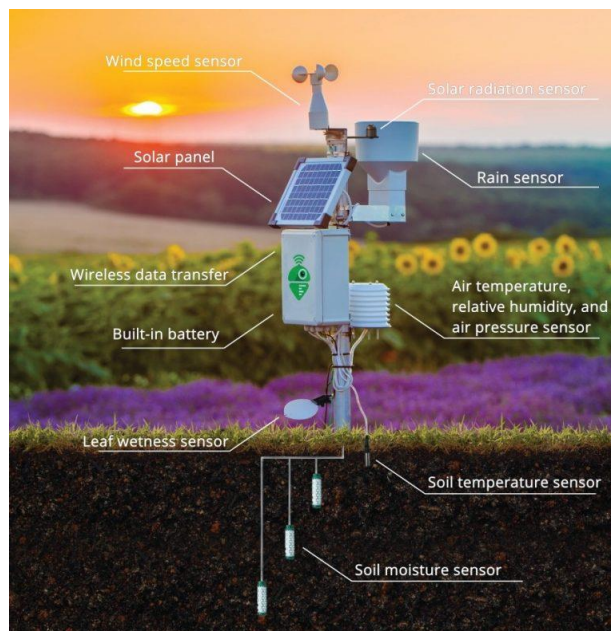


Figure 2.2 Technical Parameters of Auto Weather Station by Metrobot [4]

2.3 Related Technology

2.3.1 Software

2.3.1.1 C++

C++ is a new widely used computer programming language. It was developed from the C language. It is a general-purpose programming language that supports multiple programming paradigms, such as Procedural programming, Abstraction, Object-oriented programming, generic programming, and design pattern. This language has faster performance compared to other languages and is easy to use. It is popularly used in Operating Systems, Games, GUI Business, Web Browsers, Embedded Systems, Banking Applications, Compilers, Database Managements Softwares, Cloud/Distributed Systems, Libraries, Switches, Advanced Computation Graphics. In our project, C++ will use in Arduino IDE

2.3.1.2 Arduino IDE

The Arduino IDE is an open-source software, which is used to write sketch code, verify/compile, and upload code to microcontroller on the Arduino boards. The Arduino can show the data in both numeric data and line graphs. At first Arduino IDE can be used with Arduino boards only, but now it can be used with other boards and Node MCU. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++.

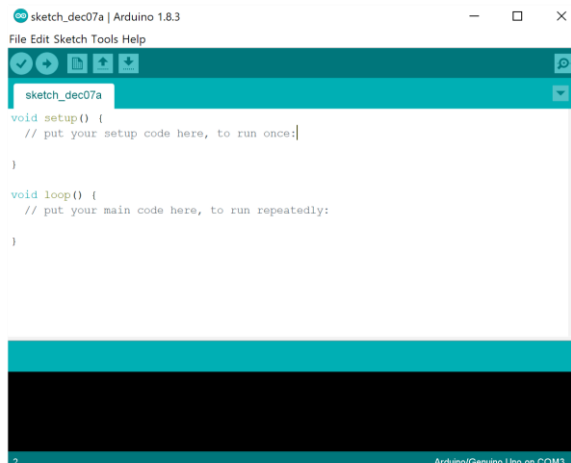


Figure 2.3 Arduino Uno Interface

2.3.1.3 Javascript

JavaScript is a text-based programming language used both on the client-side and server-side that allows you to make web pages interactive. The most popular runtime system for this usage is **Node.js**. It is mainly used for web-based applications, web browsers, software beyond the web, servers and embedded hardware controls. The example things that JavaScript is used for are web and mobile applications, adding interactive behavior to web pages, Building web servers and developing server applications, and development games.

2.3.1.4 Go or Golang [5]

Go or Golang is Open-source programming language, developed by Google in 2007, the advantages of this language is high performance as other languages, easy to learn and use. It is easy to do the concurrent programming, designed for doing Multi-Threading and Distributed System easier. Go can be used in Web Developing, Cloud Computing Systems, Command Line Interface and more.



Figure 2.4 Go or Golang Logo, Programming language [6]

2.3.1.5 Grafana [7]

Grafana is a multi-platform open source to do analytics and visualization web applications developed by Grafana Labs. It can make a dashboard containing graphs, charts, and have alerts when connected to data sources. Grafana can work with both front end and back end, written in TypeScript and Go Language. Grafana is a popular visualization tool, often used with time series databases. The original user interface of Grafana was based on Kibana version 3.



Figure 2.5 Example of Grafana Dashboard [8]

2.3.1.6 MySQL

MySQL is an open-source relational database management system (Relational Database Management System) on Structured Query Language. A relational database organizes data into one or more data tables in which data types may be related to each other; these relations help structure the data. SQL is a language programmers use to create, modify and extract data from the relational database, as well as control user access to the database. In addition to relational databases and SQL, an RDBMS like MySQL works with an operating system to implement a relational database in a computer's storage system, allows for network access and facilitates testing database integrity and creation of backups.

In our project, we choose to use MySQL because we have experience with it before, but we try comparing MySQL with others: Firebase and MongoDB

Comparing: MySQL vs Firebase vs MongoDB

	Firebase	MongoDB	MySQL
Free software			✓
Easy to use	✓	✓	✓
No SQL	✓	✓	
Easy to manage			✓
Real time database	✓	✓	✓
Open Source	✓	✓	✓
Language	C++, Java, JavaScript, JavaScript/Node.js, Objective-C, and Swift	C++, Java, PHP	Can use with mostly language

Table 2.1 Comparing MySQL vs Firebase vs MongoDB

Which Database Is Right For Your Business?

	MySQL	MongoDB
Use Case	Legacy applications or applications that require multi-row transactions (i.e. accounting systems)	Real-time analytics, content management, internet of things, mobile apps.
Data Structure	Structured data with clear schema.	No schema definition required.
Risk	Risk of SQL injection attacks.	Less risk of attack due to design.
Analysis	A great choice if you have structured data and need a traditional relational database.	A great choice if you have unstructured and/or structured data with the potential for rapid growth.

Table 2.2 Comparing MySQL vs MongoDB [9]

	Firestore	MySQL
Short Description	App development platform from Google.	Open-Source SQL Database
Price	Free to Start Pay as you go thereafter	Free to download
Open-Source	No	Yes
Vendor Lock-In	Yes	No
Managed Service	Yes	Managed hosting available on Oracle, AWS, Digital Ocean,, etc.
Data Handling	Firestore effectively handles large sets of data. It uses wide-column, key-value, document stores, or graphs and has dynamic schemas to facilitate unstructured data.	MySQL is table-based and has pre-defined schemas. It is a preferred choice for handling complex data.
Architecture	Firestore is a NoSQL database that syncs and stores data in real-time.	MySQL is an open source relational database management system that is based on SQL, the domain-specific language.
Language support	Far less vs. MySQL.	The programming languages supported by MySQL are far more than what Firestore supports. The languages include Python, C++, Ada, and so on.
Scalability	Firestore scales horizontally.	MySQL scales vertically.

Table 2.3 Comparing Firestore vs MySQL [10]

2.4.2 Hardware

Information	Picture
2.4.2.1 DHT11 Digital Temperature and Humidity Sensor [11] The DHT11 is a basic, ultra low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air, and spits out a digital signal on the data pin (no analog input pins needed). It's fairly simple to use, but requires careful timing to grab data. You can get new data from it once every 2 seconds, so when using the library from Adafruit, sensor readings can be up to 2 seconds old. Comes with a 4.7K or 10K resistor, which you will want to use as a pullup from the data pin to VCC. Specifications: • 3 to 5V power and I/O • 2.5mA max current use during conversion (while requesting data) • Good for 20-80% humidity readings with 5% accuracy • Good for 0-50 °C temperature readings +- 2 °C accuracy • No more than 1 Hz sampling rate (once every second) • Body size 15.5mm x 12mm x 5.5mm • 4 pins with 0.1" spacing • Adafruit Learning Documentation for DHTxx Sensors • RoHS compliant https://shop.evilmadscientist.com/products/mini/716	 Figure 2.6 DHT11 Picture

2.4.2.2 ESP32-CAM and OV2640 [12]

The ESP32-CAM is a very small camera module with the ESP32-S chip that costs approximately \$10. Besides the OV2640 camera, and several GPIOs to connect peripherals, it also features a microSD card slot that can be useful to store images taken with the camera or to store files to serve to clients.

The ESP32-CAM doesn't come with a USB connector, so you need an FTDI programmer to upload code through the U0R and U0T pins (serial pins).

Here is a list with the ESP32-CAM features:

- The smallest 802.11b/g/n Wi-Fi BT SoC module
- Low power 32-bit CPU, can also serve the application processor
- Up to 160MHz clock speed, summary computing power up to 600 DMIPS
- Built-in 520 KB SRAM, external 4MB SRAM
- Supports UART/SPI/I2C/PWM/ADC/DAC
- Support OV2640 and OV7670 cameras, built-in flash lamp
- Support image WiFi upload
- Support TF card
- Supports multiple sleep modes
- Embedded Lwip and FreeRTOS
- Supports STA/AP/STA+AP operation mode
- Support Smart Config/AirKiss technology
- Support for serial port local and remote firmware upgrades (FOTA)

<https://randomnerdtutorials.com/esp32-cam-video-streaming-face-recognition-arduino-ide/>



Figure 2.7 ESP32-CAM and OV2640 Picture



Figure 2.8 OV2640 Picture

2.4.2.3 Capacitive Soil Moisture Sensor [13]

This is an analog capacitive soil moisture sensor which measures soil moisture levels by capacitive sensing, i.e capacitance is varied on the basis of water content present in the soil. The capacitance is converted into voltage level basically from 1.2V to 3.0V maximum. The advantage of Capacitive Soil Moisture Sensor is that they are made of a corrosion-resistant material giving it a long service life.

- Supports 3-Pin Sensor interface
- Analog output
- Operating Voltage: DC 3.3-5.5V
- Output Voltage: DC 0-3.0V
- Interface: PH2.0-3P
- Size: 99x16mm/3.9x0.63"

<https://how2electronics.com/interface-capacitive-soil-moisture-sensor-arduino/>



Figure 2.9 Capacitive Soil Moisture Sensor Picture

2.4.2.4 NPK and pH Soil Sensor [14]

Sensor measures soil quality, NPK detect for arduino sensor. It can read NPK and pH values. Can connect directly to Arduino, connect Analog Pin of Arduino to the positive pole of the screen and connect the negative pole of the screen to the ground. No source power needed. It have switch to select if wanting to measure NPK or pH

<https://www.cybertice.com/product/2705/> เซ็นเซอร์วัดความสมบูรณ์ของดิน-npk-สำหรับ-arduino



Figure 2.10 NPK and pH Soil Sensor

CHAPTER 3

ANALYSIS AND DESIGN

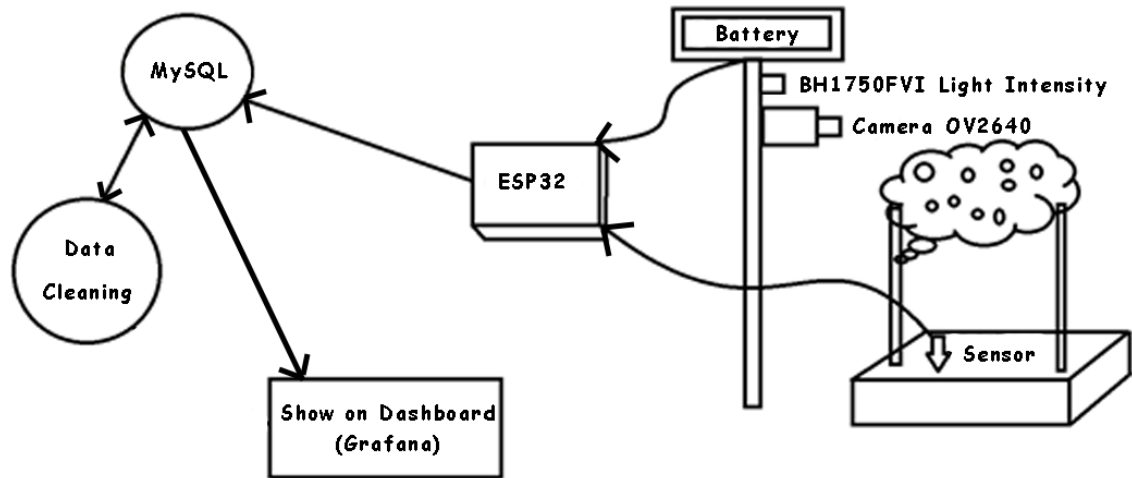


Figure 3.1 Design Draft

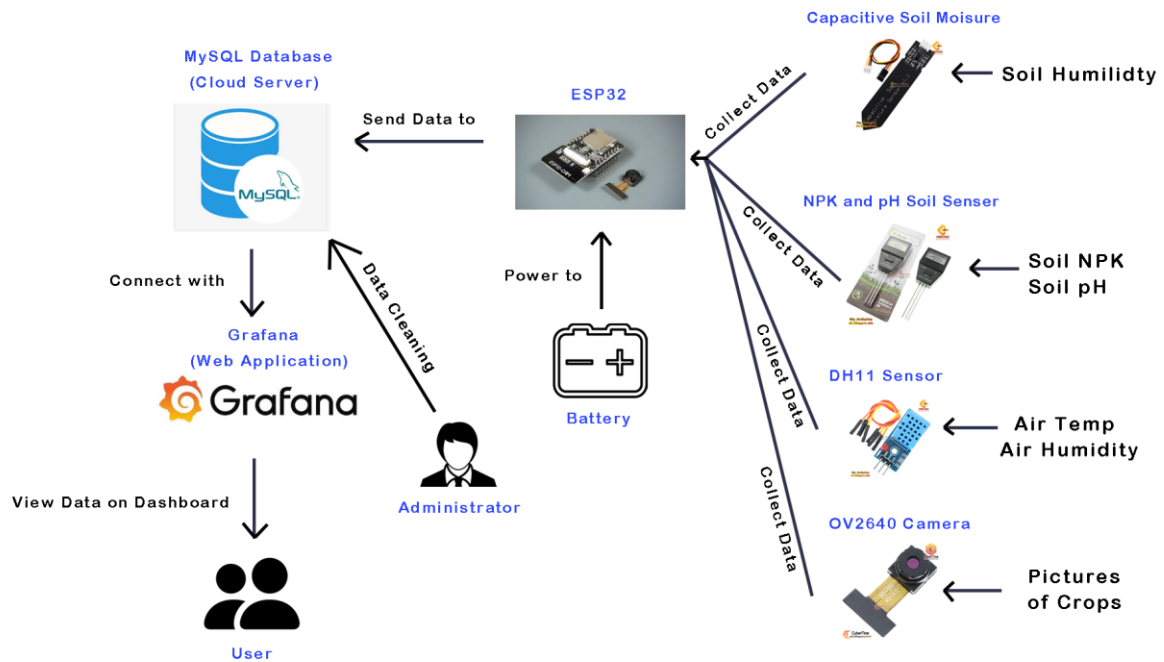


Figure 3.2 Use-case Diagram

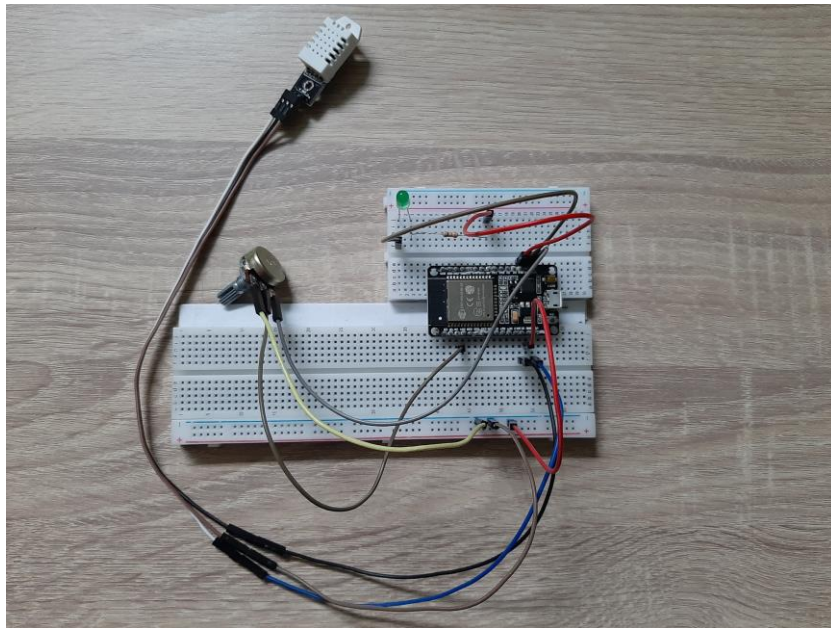


Figure 3.3 ESP32 connect with DHT22 (Prototype Testing)

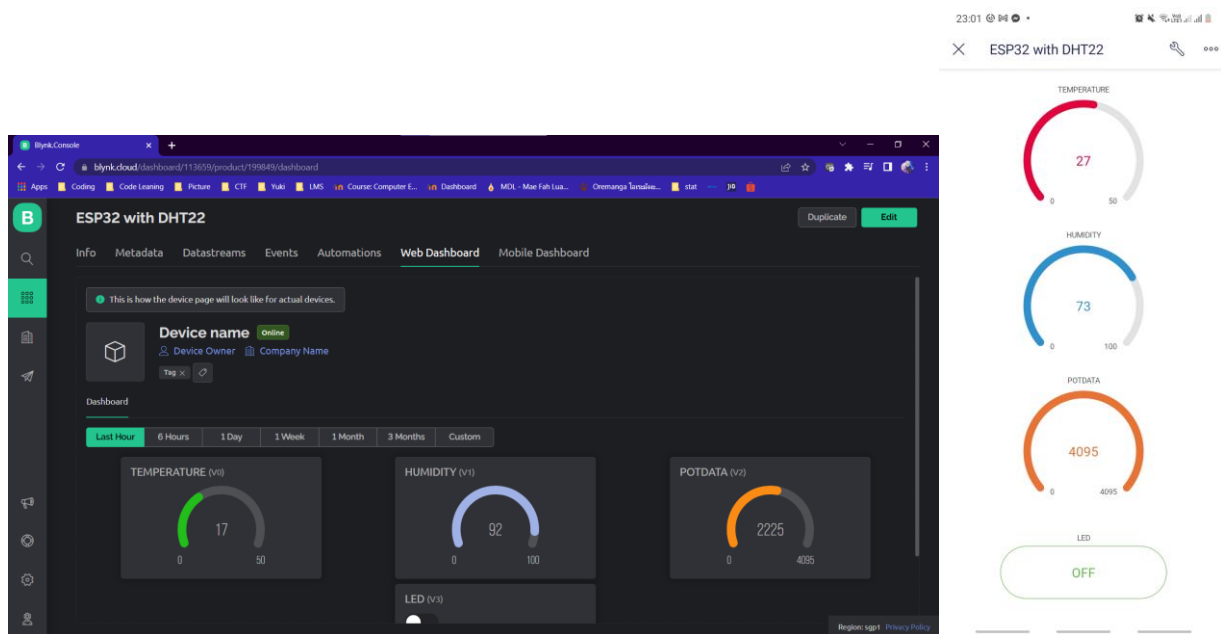


Figure 3.4 and Figure 3.5 Try ESP32 (Prototype) connect with Blynk on PC and Smartphone

REFERENCES

- [1] “Example Figure of Cloud based event in Smart Farm” [Online]. Available: https://www.researchgate.net/figure/The-cyber-physical-management-cycle-of-Smart-Farming-enhanced-by-cloud-based-event-and_fig3_327572166 [Accessed: 22-May-2022].
- [2] “Automated Greenhouse” [Online]. Available: <https://www.iaacblog.com/programs/automated-greenhouse/> [Accessed: 22-May-2022].
- [3] “Automatic Weather Station” [Online]. Available: <https://aeronsystems.com/iot/automatic-weather-station/> [Accessed: 22-May-2022].
- [4] “Technical Parameters of Metrobot Weather Station” [Online]. Available: <https://meteobot.com/en/weather-stations/> [Accessed: 22-May-2022].
- [5] “Golang คืออะไร? ดียังไง? ทำไมคนใช้เยอะ? รวมสิ่งที่ควรรู้เกี่ยวกับ Golang” [Online]. Available: <https://blog.skooldio.com/what-is-golang/> [Accessed: 22-May-2022].
- [6] “GoLang Has Finally Come of Age. Should You Jump on the Bandwagon?” [Online]. Available: <https://www.nixsolutions.com/blog/golang-finally-come-age-jump-bandwagon/> [Accessed: 22-May-2022].
- [7] “Grafana - Wikipedia” [Online]. Available: <https://en.wikipedia.org/wiki/Grafana> [Accessed: 22-May-2022].
- [8] “Grafana – An Open Source Software for Analytics and Monitoring” [Online]. Available: <https://www.tecmint.com/install-grafana-analytics-in-centos-ubuntu-debian/> [Accessed: 22-May-2022].
- [9] “MongoDB vs MySQL: Which Database is Better For Your Business” [Online]. Available: <https://appinventiv.com/blog/mysql-vs-mongodb-the-best-database-model/>. [Accessed: 22-May-2022].
- [10] “Firebase Database vs. MySQL” [Online]. Available: <https://blog.back4app.com/firebase-database-vs-mysql/>. [Accessed: 22-May-2022].
- [11] “DHT11 temperature-humidity sensor” [Online]. Available: <https://shop.evilmadscientist.com/productsmenu/716>. [Accessed: 22-May-2022].
- [12] “ESP32-CAM Video Streaming and Face Recognition with Arduino IDE” [Online]. Available: <https://randomnerdtutorials.com/esp32-cam-video-streaming-face-recognition-arduino-ide/>. [Accessed: 22-May-2022].

[13] “Interface Capacitive Soil Moisture Sensor v1.2 with Arduino” [Online]. Available: <https://how2electronics.com/interface-capacitive-soil-moisture-sensor-arduino/>. [Accessed: 22-May-2022].

[14] “เซ็นเซอร์วัดความสมบูรณ์ของดิน NPK สำหรับ Arduino” [Online]. Available: <https://www.cybertice.com/product/2705/เซ็นเซอร์วัดความสมบูรณ์ของดิน-npk-สำหรับ-arduino>. [Accessed: 22-May-2022].